

Electric Vehicle Charging Behavior Prediction using Machine Learning Models

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Abstract—The global spread of the COVID-19 pandemic has significantly impacted the electric vehicle (EV) industry. The lockdown restriction has resulted in a significant shift in the use of public charging infrastructures. This paper investigates the effects of COVID-19 on electric vehicle users' charging behavior before, after, and during COVID-19 lockdown restrictions, using the data from a public charging facility from the City of California. In this study, we performed data visualization using K-means and hierarchical clustering analysis. This work uses the vehicle's connection and disconnection time to identify common charging pattern identification and charging behavior where K-means clustering outperforms the hierarchical clustering for all three different scenarios modelled. In addition, prediction of collective charging session duration is achieved using Machine Learning Models, Random Forest and XgBoost. We achieved a mean absolute percentage error (MAPE) of 0.146 and 0.151 percent for XgBoost and Random Forest respectively.

Index Terms—Electric Vehicle; Charging Station; Machine Learning

I. INTRODUCTION

Electric Vehicles are promising the goal of sustainability in today's world. As the market for EVs is expanding, the need for the expansion of public charging infrastructures also increasing. In addition, the users perspective regarding EVs has shifted positively due, in part, to the extension of EV's battery range with each new model year. This trend has encouraged the EV owners to utilize more public charging options in addition to home charging. Thus, the EVs widespread adoption requires significantly larger deployment of public charging infrastructures. To ensure that the infrastructure satisfies the needs and interests of EV users, it is important to understand what the user behavior looks like. Although the number of vehicles using the charging station was significantly reduced due to lockdown restrictions, the charging activity has started to resurge again since April 2021 [1]. According to the International Energy Agency (IEA), in the year 2021, United States EV sales have doubled, relative to the first quarter of 2020 [1]. This significant increase of EV sales during the pandemic is a good step in the global push towards sustainability. However, as the charging pattern of EVs changes, it is necessary to understand charging behavior in public charging spaces. Thus, this study focuses on the charging behavior of users and time

spent on charging at public charging infrastructures.

This paper is organized as follows: Section II details the background and related works in charging station planning and prediction. Data used in our study and its clustering methods are given in Section III. Section IV presents the Machine Learning experimental results and discussion followed by a conclusion in Section 5.

II. BACKGROUND AND RELATED WORK

The convenience of EV drivers must be considered when planning charging infrastructure, such as charging station accessibility and waiting time for charging. In recent years, data-driven approaches that use machine learning (ML) are becoming popular to solve charging infrastructure planning problems as well as the utilization of charging. ML is also being used to ensure the convenience of users are met. Existing research works have developed various approaches to the problem of user behavior prediction for EVs. Shahriar et al. utilized K-means, hierarchical, and Gaussian mixture models to find clusters of charging behavior during the pandemic [2]. This study's clustering algorithms have identified three groups of charging behavior during the pandemic. Similarly, Wen et al. conducted a study on the impacts of COVID-19 to the Chinese EVs industry [3].

In terms of analysis and prediction using clustering techniques, Helmus et al. have found 13 distinct charging session types using the Gaussian mixture model and partitioned them to clusters of daytime and overnight charging sessions to study the duration and behavior [4], while, Gerossier et al. have utilized information from individual meters at charging stations and identified four main clusters of charging habits using hierarchical clustering [5]. On the other hand, Khan et al. used machine learning (ML) and Deep Learning (DL) approaches to predict charging behavior to infer the difference in charging performance [6]. To better understand the charging behaviors of EV users, Patrick et al. conducted statistical analysis such as Analysis of Variance (ANOVA) and other statistical tests to evaluate on charge event data recorded by data loggers located in charge points throughout Ireland [7]. Similarly, Viswanathan et al. developed an assessment model to predict the utilization of public charging infrastructure [8]. Almaghrebi et al. used Gradient Boosting (XgBoost), Random

Forest (RF), and Support Vector Machine (SVM) methods to predict the charging demand for the EV users [9]. To improve the EV Charging experience, Sun et al. used DL methods in predicting the EV charging behavior [10]. While Tianyu Hu et al. proposed DL-based method for short-term probabilistic EV charging demand prognostics, which forecasted the quantiles of future charging demand of a charging station 5 minutes ahead [11]. Jia et al. reviewed the charging behavior of EVs from data model and control perspectives [12].

To assess the impact of the pandemic, Kanda et al. conducted a study on the economic effects of COVID-19 and how significantly it reduced EV sales in the short-term. They also studied the opportunities it could provide for a sustainability transition research on mobility [13]. In order to optimize the EV charging schedule, Chung et al. used an ensemble ML algorithm to predict the EV user behavior [14]. The combination of the three algorithms: SVR, RF and diffusion-based kernel density estimator (DKDE) improved prediction performance where SVR is effective at forecasting the stay duration of EVs, while RF is better at estimating energy use. Xiong et al. developed a novel approach which combines K-Means clustering and multi-layer perceptron to model the EV user behaviors [15]. Their proposed technique was used for various kinds of charging control scheduling. Similarly, Wang et al. proposed an event-based EV scheduling algorithm to evaluate the random EV charging behavior to estimate the energy consumption and stay duration [16]. For implementing a smart EV charging algorithm to control the EV charging infrastructure with respect to users' price preferences, Wang et al. predicted the day-ahead charging load data using Auto Regressive Integrated Moving Average (ARIMA) model to determine EV charging price [17]. In order to study the real time thought process of EV users and their behavior towards charging the batteries by EV users, Ralf et al. conducted a methodological survey among the drivers from different parts of Germany [18]. Their results indicate that Battery Electric Vehicle (BEV) users tend to recharge the consumed quantities immediately due to range anxiety. Youssef et al. studied the distribution of charging session using heat maps based on the connection duration of each session. A clustering technique was also used to assess the smart charging potential of a public charging station in the Netherlands [19]. Their results show that nearly half of the overall charging station utilization occurs during the evening hours. Similarly, the length of a charging session at public charging stations was studied by Rick et al. and their results showed that the connection time exceeds the time needed to recharge the electric vehicle's batteries. Hence the charging session duration prediction plays a crucial role in determining the duration information [20]. As their study suggest, it is not only reliant on the charging duration but also on the occupancy of a particular parking lot for a longer duration which may affect the other incoming users' convenience. Sadeghianpourhamami et al. studied car arrival and departure times using three behavioral clustering analysis using Density Based Spatial Clustering of Application with Noise (DBSCAN) method for different scenarios such

as charging at home [21]. Their results show the comparison between each of these scenarios and park to charge is the largest among all three clusters. In terms of the Statistical analysis, Marco et al. have utilized a real time charging station dataset and implemented a Beta Mixture Model approach to determine the connected, idle and charge times [22]. Their result show that 50 percent of the recharge lasts for less than four hours. This helps in determining the grid infrastructure impact and EV charging management. In summary, a change to the charging infrastructure will lead to analyzing and predicting the hours spent by EV users in terms of charging duration, which greatly helps the efficiency and management of the infrastructure.

III. DATA DESCRIPTION

A. Data Collection and Preprocessing

This study used the adaptive charging network (ACN) dataset, which included over 30,000 charging sessions [23]. Real-time data was captured from charging stations located in Pasadena, California. The dataset includes logged charging events as well as user-related attributes. The study takes charging events from April 2018 to September 2021 into account, allowing the modelling of charging patterns before, during, and after the lockdown restrictions. The length of each charging session is determined by converting the connection and disconnection time as a suitable time series scale and turning each session to consider the charging duration in minutes scale. Table 1 represents the important data fields and its description utilized in this study.

TABLE I
DATASET FIELDS AND ITS DESCRIPTION

Data Field	Description
Connection time	Time when the EV is connected
Disconnection time	Time when the EV is disconnected
kWh Delivered	Amount of Energy delivered during the session

B. Data Analysis

In terms of analysis, the goal is to figure out how the charging behavior changed prior to, after and during the period of the lockdown restrictions. The time frame of the lockdown period started when the World Health Organization (WHO) announced the outbreak of the global pandemic on March 11, 2020 [24]. To analyze the charging pattern and its shift, the number of charging sessions per month of three different years are visualized.

Figure 1. explains the dramatic decrease in charging activity between August and November 2020, there was nearly zero charging activity in terms of the number of hours during this period. Furthermore, the utilization of the charging station spiked in December 2020 due to the relaxation of lockdown restrictions. The analysis is further narrowed down to number of users per hour each year as shown in Figure 2.

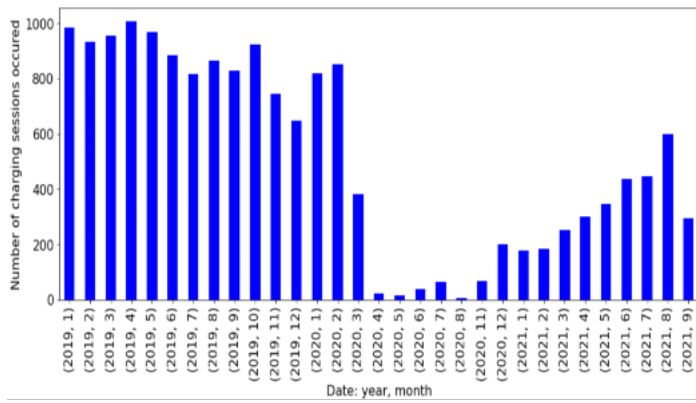


Fig. 1. Charging Sessions occurred each month from 2019-2021

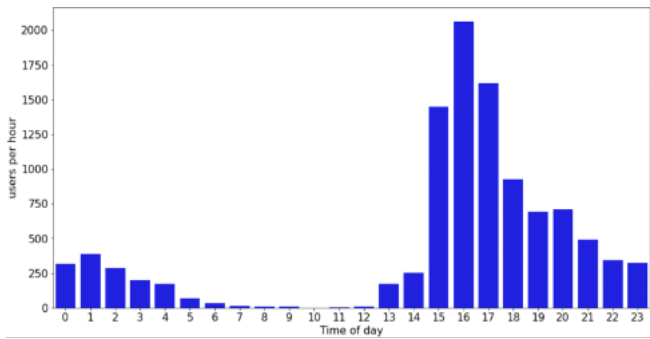


Fig. 2. Number of cars arriving at each hour in 2019

This data visualization shows the peak hours of utilization of charging stations in 2019. We see that overnight charging preferred more when compared to morning hours, and it reaches its peak by 4 p.m. With the help of this analysis, the number of users at different time frames are identified, thus contributing to the effective management of the charging infrastructure. Furthermore, to analyze the change in changing pattern in the following years, the same method is used to determine the number of users arriving at the charging station in the year 2020 and 2021 as represented in figure 3 and 4.

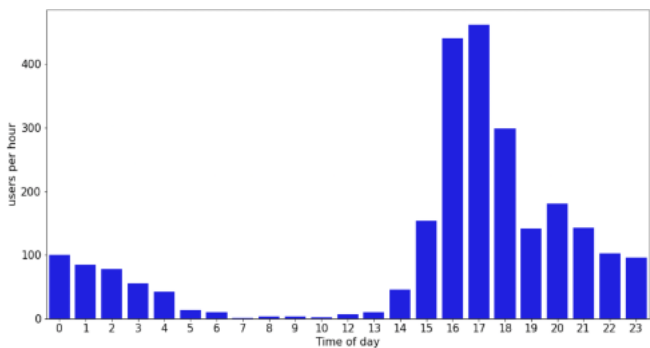


Fig. 3. Number of cars arriving at each hour in 2020

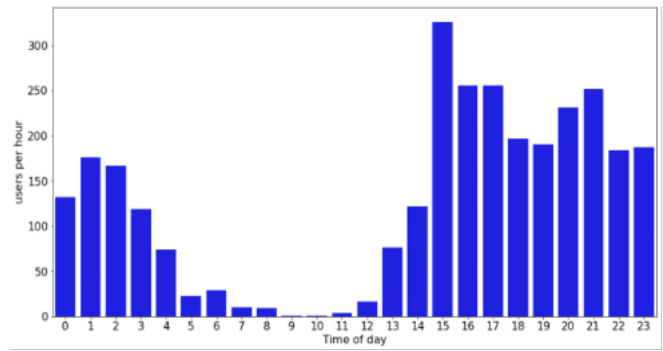


Fig. 4. Number of cars arriving at each hour in 2021

C. Clustering Techniques

K-means and hierarchical model, two popular clustering algorithms are used to find common charging behavior. To determine the optimal number of clusters, before training the datasets, an elbow plot is followed. Figure 5 shows the elbow plot varying the number of clusters from 1 to 10 with the cluster Sum of Squared Error (SSE). As a result, the optimal number of clusters are set to three for both clustering algorithms.

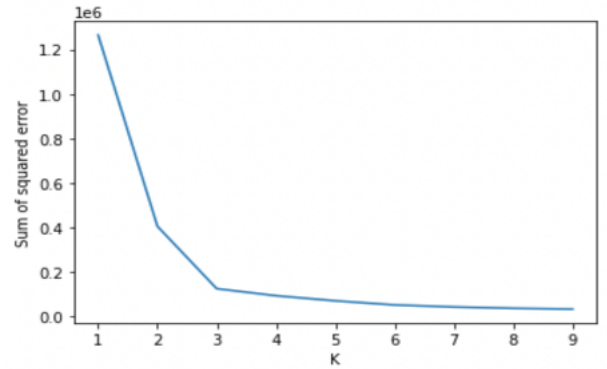


Fig. 5. Elbow plot for deciding the number of clusters

In K-means clustering, data points are assigned to k-centroid points at random. This method picks the right values for initial centroid, called K-means seeding. The number of clusters is denoted by the letter K. The data points are then assigned to new centroids in an iterative process based on their similarity [25]. The centroids are computed in the meantime, and each iteration is updated. The procedure is repeated until the algorithm converges and the cluster labels remain unchanged as shown in figure 6. Each charging session from 2019, 2020, and 2021 is considered separately to implement in K-means and hierarchical clustering algorithms. To perform the clustering, the charging behaviors were identified by selecting the connection and disconnection times. The clustering was done in Python using the scikit-learn library [26].

In Figure 7. Cluster zero represents the overnight users utilizing the charging stations from 12AM Cluster one represents the overlap between the users in the evening hours. Silhouette

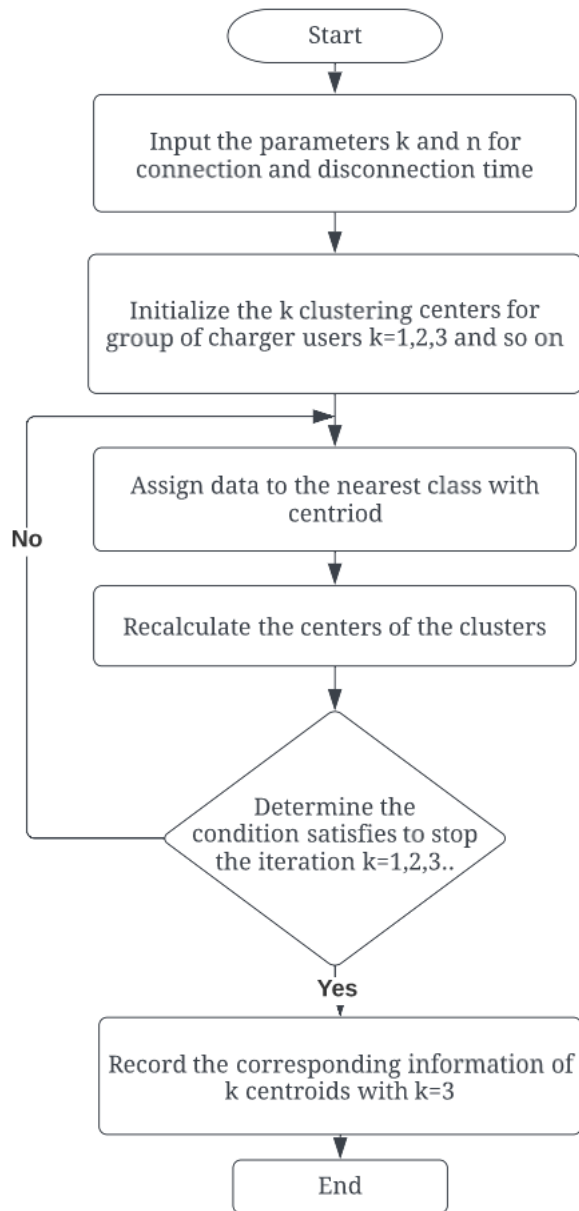


Fig. 6. Flowchart for K-means Clustering algorithm based on connection and disconnection time

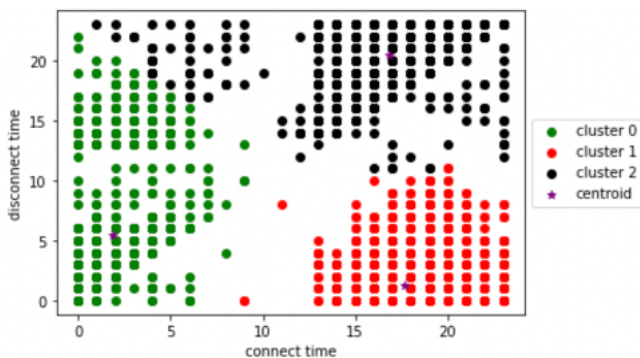


Fig. 7. K-means Clustering results of connect and disconnect time for 2019

index metrics are used to validate the quality of the clusters. The results of this clustering technique are summarized in Table 2 for different years using the silhouette indices.

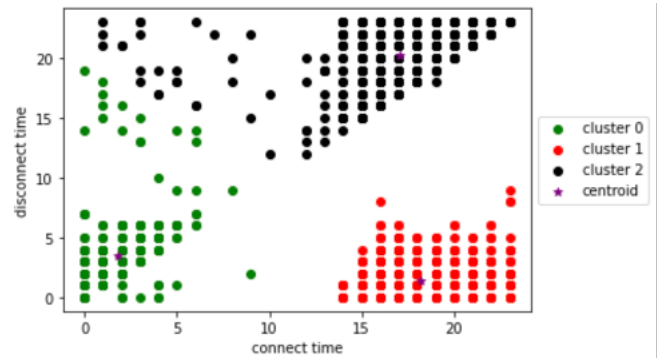


Fig. 8. K-means Clustering results of Connect and disconnect time for 2020

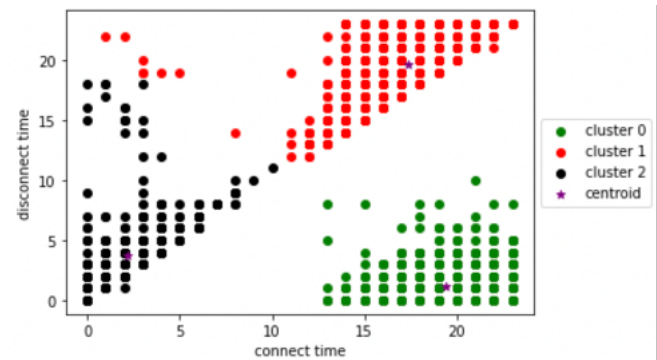


Fig. 9. K-means Clustering results of Connect and disconnect time for 2021

TABLE II
SILHOUETTE INDICES FOR CLUSTERING ALGORITHMS

ML Models	Silhouette Indices		
	2019	2020	2021
K-Means	0.7726	0.7879	0.7585
Hierarchical	0.7542	0.7839	0.7518

The silhouette index for the clusters formed using the K-means algorithm for the user connection and disconnection time were taken for 2019, 2020 and 2021. Their values are 0.7726, 0.78799 and 0.7585, respectively for figure 7 to figure 9. The hierarchical clustering algorithm produced similar number of clusters, as represented in Figure 10. By comparing the results, the K-means algorithm slightly outperformed the hierarchical clustering algorithm based on its accuracy metrics. The agglomerative hierarchical clustering algorithm for the user connection and disconnection time using Ward's method [27] is calculated with the internal validation scores of 0.766, 0.783 and 0.758 respectively.

Although the formation and the number of clusters is similar between each year, the charging station utilisation based on number of users varies by each cluster.

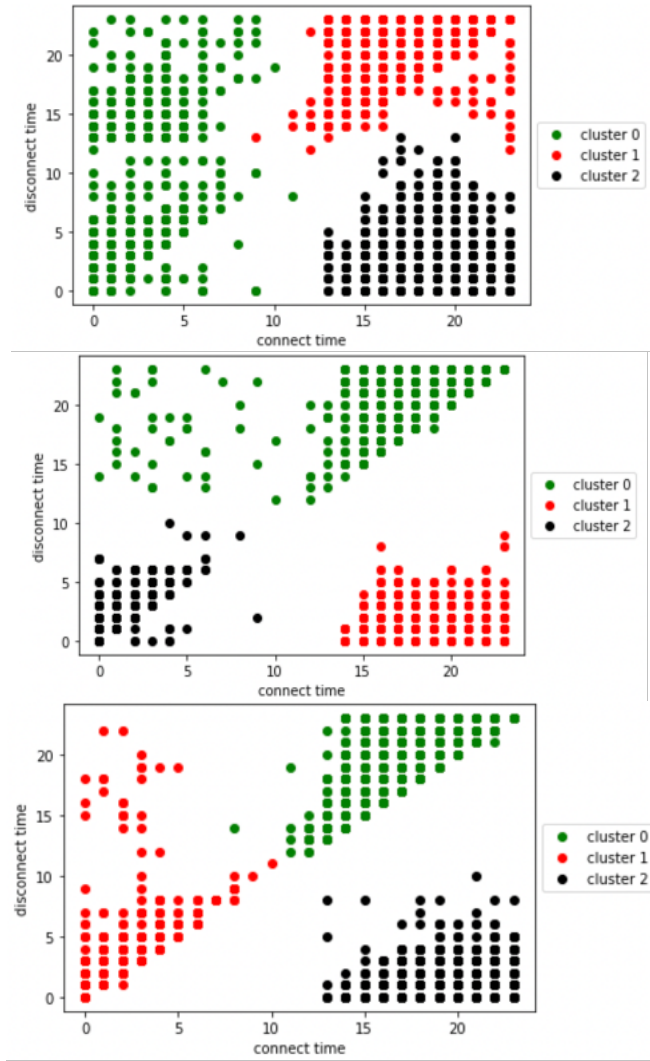


Fig. 10. Hierarchical Clustering results of Connection and disconnection times for 2019, 2020 and 2021

IV. MACHINE LEARNING MODEL RESULTS

The difference in time between connection and disconnection hours is used in the experimental analysis for predicting session duration with Machine Learning models such as Random Forest and XgBoost with different hyperparameter settings as shown in Table 3.

TABLE III
HYPERPARAMETERS OF MACHINE LEARNING MODELS

ML Models	Hyperparameters	
	Parameter	Value
RF	n-estimators	10
	Random State	10
XgBoost	n-estimators	200
	Max Depth	1

Figure 11. represents the overall session duration from 2018 to 2019. The data is obtained from the difference in connection

and disconnection hours. As the pandemic had the greatest impact on charging hours, the session duration considered is based on years from 2018 and 2019. The overall session duration is split into an 80-20 percent ratio for the purpose of training and test validation of the ML models. Session duration that occurred from the period of April 2018 to September 2019 is considered as the value to train. Test values from September to December 2019 for the purpose of testing the ML models such as Random Forest [28] and XgBoost [29]. Figure 12 represents the forecast of the session duration for the next 10 days from the test value in the validation.

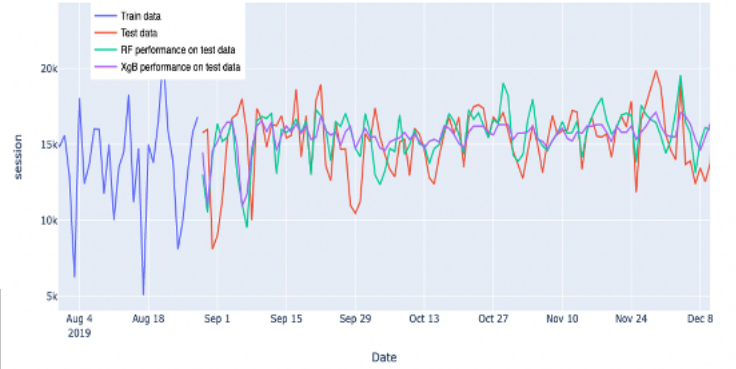


Fig. 11. Machine Learning performances over the test data of charging session duration

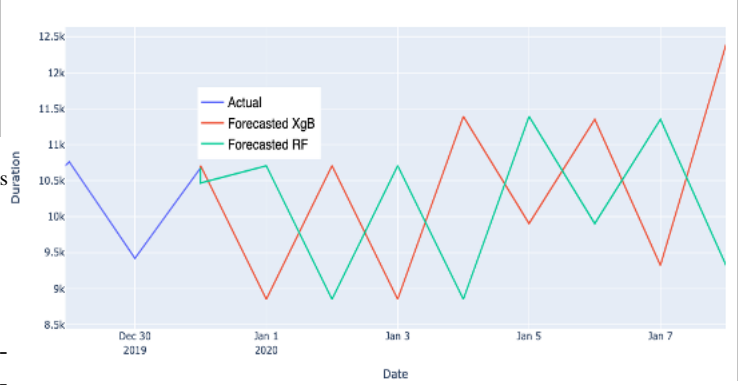


Fig. 12. Charging session duration forecast for the next 10 days

The prediction errors of the session duration is calculated for the above-mentioned ML methods using Mean Absolute Percentage Error (MAPE) scores as shown in Equation 1. The results and performance of both ML models on the dataset are then compared. Table 4 explains the comparative analysis of the predicted session duration of both ML models implemented. The proposed method reduces the forecast error, but there is still further room for improvement.

$$MAPE = \frac{1}{n} \sum_{i=1}^D (|e_t|) / d_t \quad (1)$$

TABLE IV
RESULTS OF ML MODEL

ML Models	MAPE (Percentage)	
	2018-2019	2021
RF	0.15169	0.19642
XgBoost	0.14630	0.17800

V. CONCLUSION

In this work, we utilised the unsupervised clustering techniques to examine the charging pattern before, after, and during lockdown restrictions. According to the analysis, there is a noticeable shift in time of utilization of charging stations, because of the lockdown measures. There were no charging sessions between the beginning of August and the middle of November 2020, but a significant increase was seen starting in the middle of the year 2021. If this uncertainty in charging behavior continues, charging station usage will fluctuate on a regular basis, potentially affecting charging station planning and user convenience. In addition, in a rapidly economically growing state like California, the number of users is likely going to rise in the near future. As a result, this work is being provided in order to identify a common pattern among users of charging stations, as well as to predicting each charging duration, which is critical for charging station development.

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